Efficiency and acceleration in the accumulation of children's early vocabulary

Michael C. Frank

Abstract

Children learn hundreds of words over the first few years of their life. This process begins slowly but increases until only a few exposures to a word are necessary for a child to learn it. Yet there are also wide variations between children, and the nature of this variation has not been accurately characterized. Here we present a statistical model describing this process as accelerating accumulation, with children varying in both the slope and intercept of their accumulation. Using data from three languages, this model provides a good characterization of longitudinal growth and gives insights into the sources of age and socioeconomic variation. Using dense observations of children's early language environment, the model also can be used to estimate how much variation between children is due to differences in language input, finding estimates that converge with recent meta-analyses. In contrast to children, large language models are not well-described by this model: they show standard accumulator dynamics and very limited variability.

Children's early vocabulary grows very rapidly during early childhood, with large variation between children (Fenson et al. 1994, 2007; Frank et al. 2021). The growth has theoretical, practical, and methodological importance. From a theoretical perspective, growth mechanisms shed light on children's fundamental learning mechanisms and how they change across childhood (McMurray 2007; Bloom 2002; Frank et al. 2021). From a practical perspective, early vocabulary is an important precursor – and predictor – of academic achievement [Marchman and Fernald (2008);CITES]. From a methodological perspective, vocabulary also affords a unique psychometric opportunity to study human variation because, unlike nearly all important human psychological metrics, there is a true zero point – a time before which children know no words. Thus, absolute levels of variability can be calculated.

One dominant viewpoint is that children's vocabulary growth can be characterized as a process of accumulation (McMurray 2007; Mitchell and McMurray 2009; Hidaka 2013; Mollica and Piantadosi 2017; Kachergis, Marchman, and Frank 2022). Each distinct word (type) can be conceptualized as a "bucket," and every token of language input then is a "drop" that falls into its respective bucket. When each bucket is full, the word is learned (McMurray 2007; Kachergis, Marchman, and Frank 2022). These models have been used to recover interpretable differences in "bucket size," often based on word properties including concreteness and syntactic category (Hidaka 2013; Kachergis, Marchman, and Frank 2022). They also provide a graded explanation for the widely-observed "vocabulary spurt" phenomenon – in which children's rate of vocabulary learning appears to increase dramatically (Frank et al. 2021; Ganger and Brent 2004; Gómez Díaz et al. 2024).

The accumulator view connects with a viewpoint in which the quantity of language that children hear plays a strong causal role in the speed of their learning (Fernald and Marchman 2012). Indeed, there is systematic evidence that variation in children's input is related to variation in vocabulary [Fernald and Marchman (2012);Weisleder and Fernald (2013);CITES]. Studies of input have identified systematic variation in *quantity* in terms of sheer amount of tokens as well as variation in *quality*. Measuring what makes input high quality has been challenging, but higher quality input likely has higher lexical diversity, longer sentences, more declarative and interrogative content relative to imperatives, and more elaborated discourse [LOTS OF CITES].

A naïve interpretation of the accumulator view would suggest that most of the variance in vocabulary growth should be related to input. Two recent meta-analyses complicate this view, however. Anderson et al. (2021) showed that only a modest proportion of the variance in vocabulary outcomes is related to input quantity (XYZ) and quality (XYZ). A newer meta-analysis of quantity specifically confirms and extends the proportion of variance due to quantity to 4–7% of total variation in vocabulary (Coffey and Snedeker 2026). Thus – as we might have supposed based on other research on human variation – not all variation in vocabulary between children can be attributed to input differences.

As this discussion indicates, despite the promise of the accumulator viewpoint, it has not been used successfully to understand variation between individuals. Instead, the majority of applications have focused on variation between words. Further, the rapidity of children's early word learning stands as a rebuke to this view. Sometimes a small number of examples is enough to form a lasting representation (Markson and Bloom 1997; Woodward, Markman, and Fitzsimmons 1994). Perhaps only a small number of high-quality instances is sufficient for learning (Trueswell et al. 2013; Mollica and Piantadosi 2017), but if so, why do some words take much longer to learn?

The goal of this work is to address these puzzles through systematic investigation and extension of accumulator models. All models seek to explain the observation that the overall size of children's vocabulary increases supra-linearly. Does this growth imply developmental changes in the underlying process?

Following previous work, we start with the idea that children's *internal learning ability* (not just their observed vocabulary growth) accelerates across development (Hidaka 2013; Mayor and Plunkett 2010). In the bucket metaphor, the drops get bigger as children get older. We begin by extending the psychometric accumulator model of Kachergis, Marchman, and Frank (2022): our key addition is to describe variation between individual children by modeling both their individual learning efficiency and the acceleration of their learning. This extension defines a longitudinal growth model that can be fit to data, and that provides a simple reconciliation between slow accumulation and fast mapping: the number of examples of a word necessary for learning declines by multiple orders of magnitude from age one to age three.

To constrain our model, we use data from the MacArthur-Bates Communicative Development Inventories [MB-CDI; Fenson et al. (1994);Marchman, Dale, and Fenson (2021)], a popular parent report instrument in which parents are asked to indicate which words their child produces and understands. Here we focus exclusively on production, which is used with older children and which tends to be a more reliable measure (Frank et al. 2021). We make use of longitudinal data from Wordbank (Frank et al. 2017), an open repository of data from the MB-CDI.

To address questions about the relationship between language input and growth, we fit a Bayesian version of our growth model. We make use of new data resources to estimate the magnitude of linkages between input and vocabulary growth. First, we estimate plausible population values based on independent estimates of population variation in language input (Coffey et al. 2024). In addition, we use measures of individual children's language input from egocentric videos (Long et al. 2024) and day-long home recordings (Egan-Dailey and Bergelson 2025; Adams et al. 2018; Fernald, Marchman, and Weisleder 2013) to estimate the input-uptake relationship more precisely for a subsample of children.

Overall, our analysis supports a view in which children's vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. We end by comparing the fit of our models on human data to their fit on large-language models. Making use of GPT-2 models trained on human developmental data, we show that – in contrast to children – these models do appear to function as pure accumulators. This analysis exposes a critical disanalogy between LLMs and human learners and suggests a potential route for understanding the vast "data gap" between children and large language models (Frank 2023; Warstadt et al. 2023).

Model

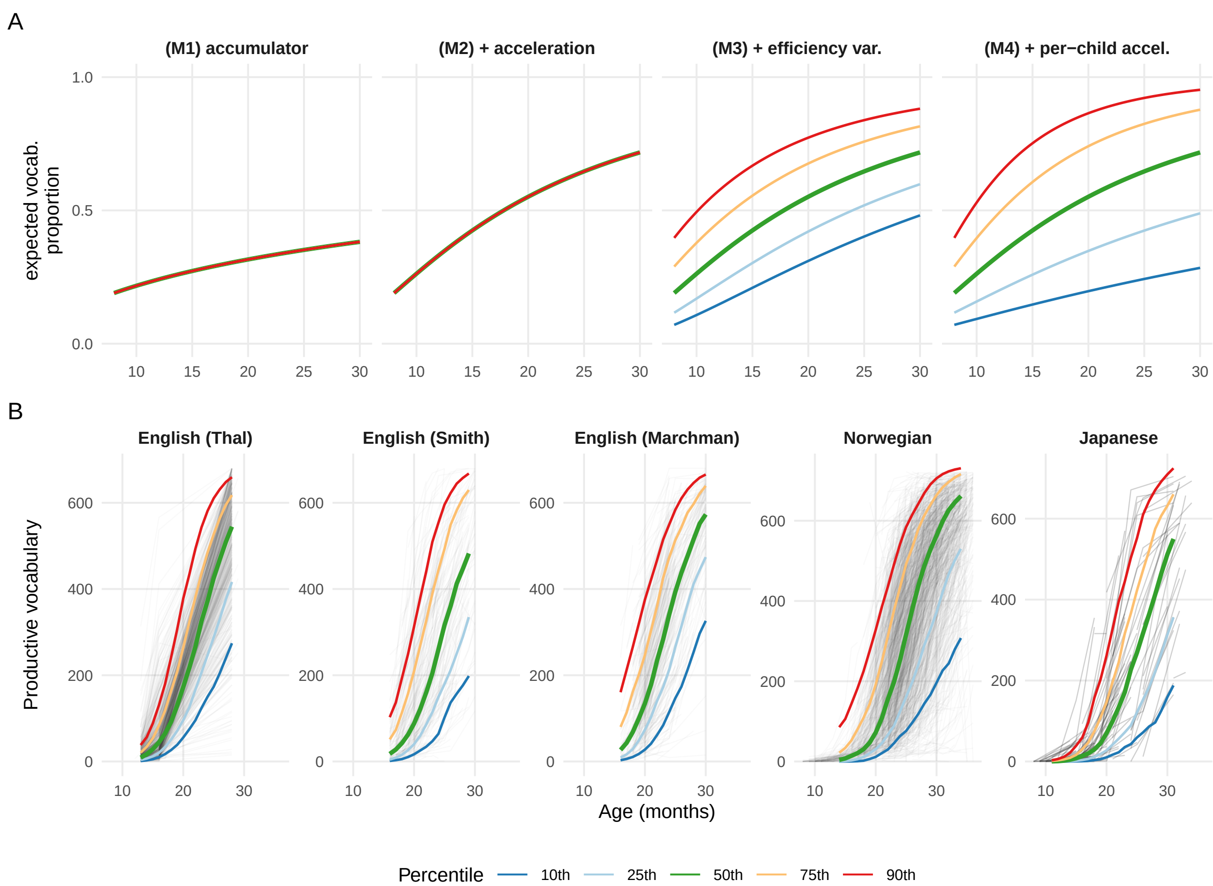

Figure 1

We present a nested set of accumulator-type models, many of which have been presented in the literature. As we show below, each refinement of the model contributes to its ability to describe the variability between children. We start with the Rasch model from psychometrics (Rasch 1960; Bond and Fox 2013), which assumes that each child i has a learning ability θ_i and each word j has a difficulty δ_j. The probability that child i produces word j is

$$P(w_{i,j} = 1 \mid \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$

Following Kachergis, Marchman, and Frank (2022), we can write our first model as a Rasch-style accumulator model (Model 1), specifying that θ_i is not a single parameter but instead a function of accumulation over time. In the simplest form:

$$\text{M1: } \theta_i(t) = \log(r_i) + \log(t/a_0)$$

where r_i is the child's rate of language input (in tokens per month) and t is the child's age (in months, anchored at a_0); note that we use log age to simulate linear accumulation because $\exp(\theta) = r \cdot t$. This form instantiates the basic hypothesis that the only thing that changes over time is more input, and the only source of variation between children is variation in rate.

We compare this model to an accelerating accumulator model (Model 2), in which the amount learned from each exposure increases with time by an exponent κ.

$$\text{M2: } \theta_i(t) = \log(r_i) + \kappa \ \log(t/a_0)$$

When $\kappa > 1$, each successive month yields more accumulation than the last.

Neither of these models has any way to capture variation across children beyond variation in input rate. Thus, we augment the basic accelerating accumulator, rewriting it as the sum of either one or two individually variable quantities:

$$\text{M3: } \theta_i(t) = \xi_i + \kappa \ \log(t/a_0)$$

$$\text{M4: } \theta_i(t) = \xi_i + \kappa_i \ \log(t/a_0)$$

The intercept term in both models $\xi_i = \log(r_i) + \log(\alpha_i)$ can then decompose individual variation into both input rate-related and other variation in children's efficiency. Similarly, in M4, the slope term $\kappa_i = \kappa + \gamma \log r_i + \zeta_i$ adds *individual variation in acceleration*, decomposing acceleration into a population term, input-related acceleration, and a child-specific residual acceleration ζ_i (the part not explained by input). The coefficient γ then captures the strength of the relationship between input and acceleration.

Model 4 instantiates the idea that children vary in their base rate of efficiency and their acceleration, creating variant of a standard longitudinal growth model in which children vary on their slope and intercept (Grimm, Ram, and Estabrook 2016). Note that, while we define the relationship of ξ_i and

κ_i to a child's input r_i, we can nevertheless fit the simplest form of the model by just allowing these parameters to vary as random child-level slopes and intercepts.

As we will show, this model provides a good description of variation in children's longitudinal growth trajectories. Figure 1 (top) shows schematic predictions from a pure accumulator model (M1), an accelerating accumulator (M2) in comparison to models including child-level variation in efficiency (M3) and both efficiency and acceleration (M4). Each model's predictions are represented as ability scores (θ) across age and then also projected into proportion correct scores on a discrete set of vocabulary items, representing an observed accuracy score that is subject to compression at floor and ceiling.

A variety of previous work has focused on estimating the properties of individual words (Goodman, Dale, and Li 2008; Hidaka 2013; Roy et al. 2015; Braginsky et al. 2019; Kachergis, Marchman, and Frank 2022; Kachergis et al. 2022). These analyses reveal that there is predictable word-level difficulty information but that only a modest portion of the variation between words is carried by frequency information. Words can be hard for a child to acquire because they vary also in the complexity and concreteness of their meanings, their syntactic category and syntactic context, their phonology, and many other factors. In the current work, we estimate word difficulty as a free parameter, and focus instead on variation between children.

Our analysis generally depends on the availability of longitudinal data at the item level. The different models described above are not distinguishable from the cross-sectional, aggregate data that are typically considered in many analyses of early vocabulary. First, aggregate data do not allow for the estimation of individual word difficulties, thus the distribution of difficulties remains latent and different difficulty distributions can be posited to explain the accelerating shape of vocabulary growth (McMurray 2007; Mitchell and McMurray 2009). Second, cross-sectional data do not allow for accurate estimation of whether variation is due to differences in efficiency (intercept) or acceleration (slope) (see Supplemental Figure 1).

Results

Children's longitudinal growth reflects differences in both efficiency and acceleration

We first retrieved longitudinal data on monolingual, typically-developing children's vocabulary from Wordbank (Frank et al. 2017), an open archive of data from the MacArthur-Bates Communicative Development Inventory and its international adaptations. Table 1 shows the four longitudinal datasets in Wordbank that contained more than > 100 children with multiple observations.

Table 1: Characteristics of the datasets.

Dataset	Citation	N	Admins (mean)	Ages	Recordings
Norwegian	@simonsen2014	1,676	1.6	8–36 mo	—
Japanese	@hagihara2023	178	1.3	8–36 mo	—
BabyView	@long2024	22	5.0	6–32 mo	Headcam (n = 5,739 videos)
SEEDLingS	@egan2025	44	12.0	6–17 mo	LENA (n = 525 recordings)
AM2018	@adams2018	66	3.0	16–24 mo	LENA (n = 126 recordings)
FMW2013	@fernald2013	51	3.0	18–30 mo	LENA (n = 51 recordings)

Taking advantage of the equivalence of Rasch models to generalized linear mixed effects models (De Boeck et al. 2011), we fit the set of models shown in Figure 1 as a series of logistic regressions. As described above, this initial set of fits do not attempt to estimate the relationship with input but instead simply attempt to describe the distribution of vocabulary growth curves. Each model predicted the probability of an individual child producing a particular word as a function of age (either unmodified in M1 or multiplied by a constant in M2) and a random effect of word difficulty. Successive models add random intercepts (M3) and slopes (M4) by child.

Model 4, which incorporated variation in acceleration and efficiency, was the best fitting model across all datasets, with AIC and BIC differences shown in Table 2. Figure 1 (bottom) shows the fit of the four model variants; the increasing spread of children's trajectories with age is especially noticeable for low-vocabulary older children [who may be classified as "late talkers"; Fernald and Marchman (2012); Rescorla (2011)].[1]

Table 2

		ΔAIC				
Model	Age	Thal	Smith	Marchman	Norwegian	Japanese
M1. accumulator	—	703,220	273,732	251,211	2,281,606	94,503
M2. + acceleration	linear	206,553	179,872	134,596	1,377,615	41,169
M2. + acceleration	log	204,930	178,471	133,877	1,348,013	39,054
M3. + efficiency var.	linear	35,856	17,418	13,752	130,625	3,850
M3. + efficiency var.	log	34,219	15,356	13,135	115,116	3,227
M4. + acceleration var.	linear	1,134	553	0	4,301	169
M4. + acceleration var.	log	0	0	30	0	0

Efficiency and acceleration relate to separate demographic predictors

Children's vocabulary is known to be related to a number of sociodemographic predictors, including sex and maternal education (which is interpreted as a proxy for socioeconomic status). In particular, female children show consistently larger vocabularies than male children across dozens of languages and countries (Eriksson et al. 2012; Frank et al. 2021). Similarly, children of more educated mothers tend to show larger vocabularies, though this effect is somewhat more heterogeneous across countries and languages (Frank et al. 2021).

Our longitudinal model offers a summary of individual children's growth in terms of both their efficiency (intercept) and their acceleration (slope). This decomposition allows us to ask whether these two demographics effects – sex and maternal education – are seen in efficiency, acceleration, or both. For each child in our analysis, we extracted the best linear unbiased prediction (BLUP) from our models. We then regressed intercept and slope terms against each demographic predictor separately for each language. Figure 2 shows the coefficients from this analysis, along with a cross-language meta-analytic estimate.

[1] Following the literature, we primarily investigate linear accumulator models, which show power-law growth of ability over time because θ is a function of $\log(t)$. However, it is also possible that ability grows linearly in time, which would lead to exponential increases in the probability of word production. We fit linear-time models, but found that they were dispreferred statistically, though differences were relatively small within the restricted age range for which we have data (δAIC 0–4,301; see Supplemental Figure 2 for model fits).

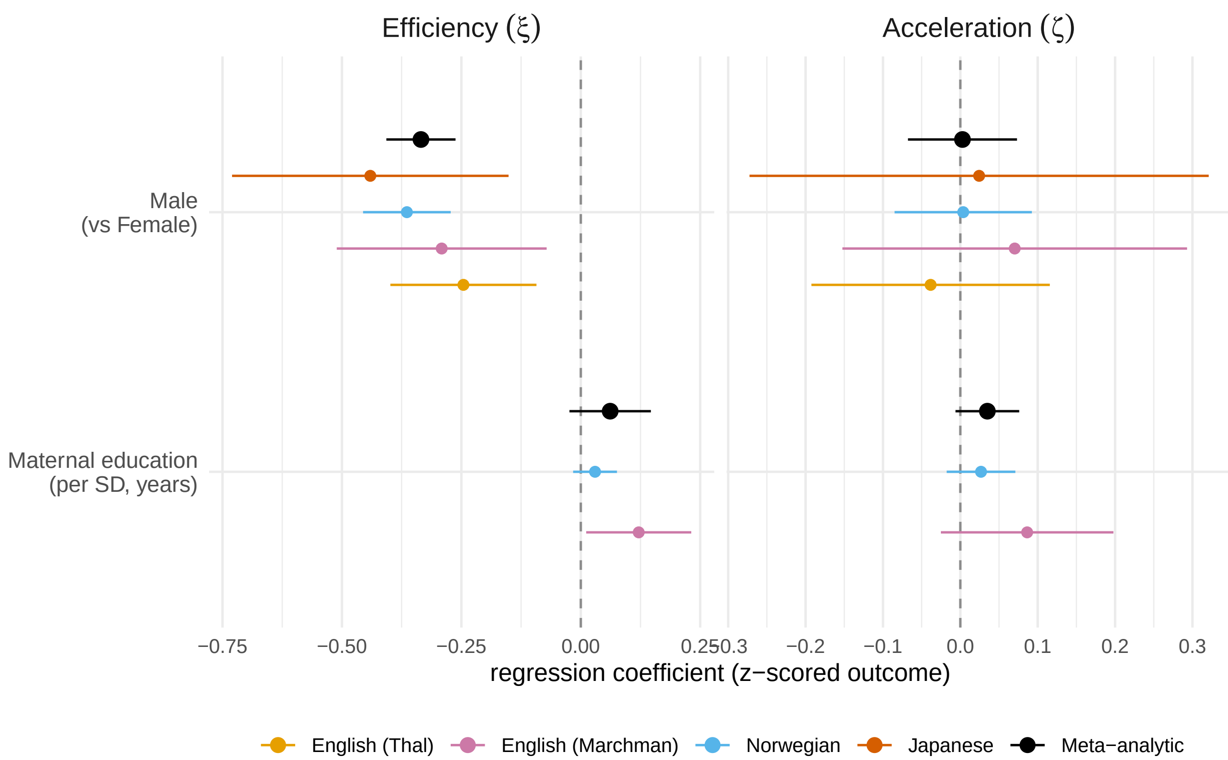

Figure 2: Regression coefficients predicting individual children's efficiency and acceleration as a function of gender and maternal education. Black dots show meta-analytic estimates from a random-effects meta analysis. Error bars show 95% confidence intervals.

Sex effects show the predicted female advantage across all four datasets, but this effect manifests itself only in efficiency ($\beta = -0.33$, $p =< .001$) not acceleration ($\beta = 0$, $p = .939$). In contrast, though maternal education data were only available for two languages, effects were numerically present in both measures, though neither was significant (efficiency: $\beta = 0.06$, $p = .157$; acceleration: $\beta = 0.03$, $p = .096$). One interpretation of these findings is that sex effects reflect a small constant offset in average ability. In contrast, socioeconomic differences as reflected in maternal education might continue to compound across development (Fernald, Marchman, and Weisleder 2013; Hart and Risley 2003).

Population input-related variation corresponds to meta-analytic estimates

In our prior model fits, we fit population variation in efficiency and acceleration without considering the connection to language input. We next consider whether this relationship can be estimated at a population level. We use markov chain monte carlo to fit the full version of model 4, with both efficiency and acceleration linked to input variation (see Supplemental Methods for full model specification). We estimate these linkages by assuming a known level of population variation in language input rate: σ_r. We can then examine the variance due to input variation vs. population by fitting the parameters $\pi_\xi = \frac{\sigma_\alpha^2}{\sigma_\alpha^2+\sigma_r^2}$ and $\pi_\kappa = \frac{\sigma_\alpha^2}{\sigma_\varsigma^2+\sigma_r^2}$, which respectively define the proportion of population variation in efficiency and acceleration due to input variation.

Estimating children's input range is notoriously tricky, since the rate of language that children hear varies across cultures, families, activities, and measurement occasions (Coffey et al. 2024). To establish a range of plausible values for σ_r, we surveyed the literature for plausible estimates [CITES;(**sperry2018?**);Coffey et al. (2024)]. In our main simulations we set $\sigma_r = .53$, a value drawn from (**sperry2018?**), who estimated a mean of 419k tokens per month ($±$1 SD: 329k – 957k). See Supplemental Methods for other input estimates.

results

[Sensitivity checks]

[relation to meta-analysis]

Input modestly predicts efficiency and acceleration for individual children

While the population estimates of input-sensitivity given above match our expectations from meta-analyses, they do not offer a direct estimate of input couplings within individual children. For this, we turn to a set of four longitudinal datasets that include dense home recordings (see (**tab-datasets?**)). Three of these include day-long audio recordings using LENA (Language Environment Analysis) recorders, which offer an automated adult word count estimate that has been validated by a number of transcription studies (**cristia2021?**). The last includes short video recordings of egocentric video recorded in children's homes, with word counts estimated from automated transcripts. Each of these provides a noisy proxy for total home input, but nevertheless allows us to estimate models that include individual input.

We fit the same model linking input estimates to variation in efficiency and acceleration, but this time with observations of r_i for each child. Results for π_alpha and π_zeta are shown in Figure 3.

RESULTS

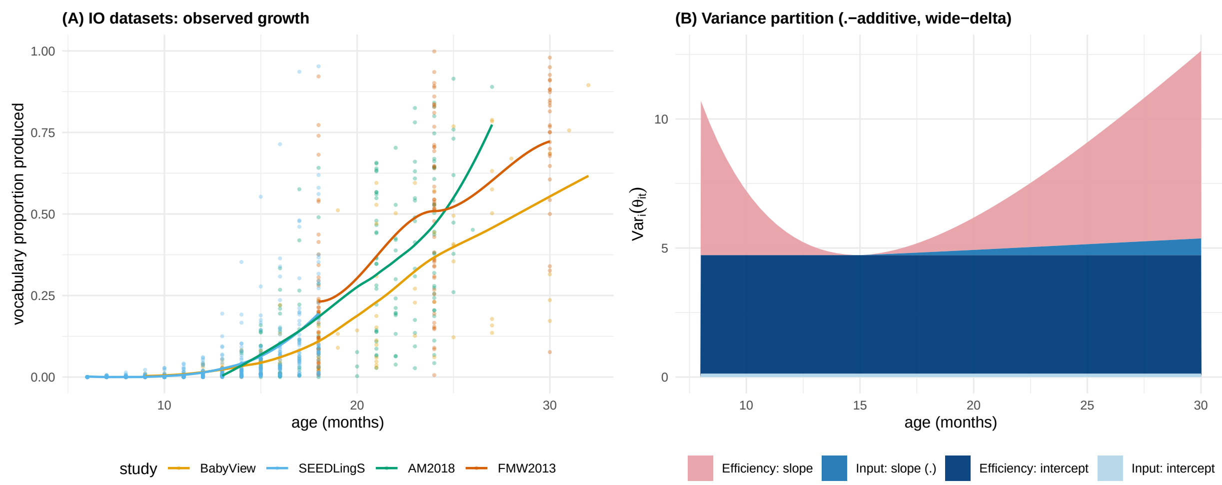

Figure 3

Processing speed is strongly associated with efficiency

Two of the datasets we included also include longitudinal reaction time data from the looking-while-listening paradigm (LWL), archived in the Peekbank repository (**zettersten2022?**). In LWL, children see two pictures (e.g. a book and a car) and their eye movements are recorded as they hear a sentence that refers to one of the pictures (e.g., "look at the book"). If the child is looking at the distractor image (the car), then their reaction time to shift their eyes to the target provides an index of their speed of word recognition (**fernald2008?**; **frank2026?**).

We fit a modification of our model to this subset of the data (N=XYZ). Efficiency and acceleration terms included *both* input and LWL reaction time as predictors.

Growth in ability leads to rapid changes in learning times for individual words

As an illustration of the changes in efficiency and acceleration that we observe in our data, we computed the average proportion of exposures necessary for a child to learn a word across ages. Figure 4 shows a model-derived estimate of the average number of times a word is heard before more than 50% of children produce it (See SI: Changes in Word Efficiency), using frequency estimates from the CHILDES Corpus (**macwhinney2000?**; Sanchez et al. 2019).

Across different word classes (e.g., nouns, verbs, closed class words), efficiency increases exponentially, such that early-learned words such as "book" may be heard tens of thousands of times before they are produced, while later learned words such as "vitamin" might only be heard hundreds times before being produced. Indeed, this analysis likely understates the degree of change in efficiency as it assumes that each exposure to a word is equally meaningful. In fact, individual exposures vary widely in how much information they give. Sometimes words are overheard rather than directed to the child, or lack physical context to allow for clear inference (**effective_learning_instances_erika?**; **trueswell2014?**; Mollica and Piantadosi 2017). Thus, in line with both intuition and experiments, the total number of exposures required on average to learn a word might indeed be only a small handful by age three (Markson and Bloom 1997; Bloom 2002).

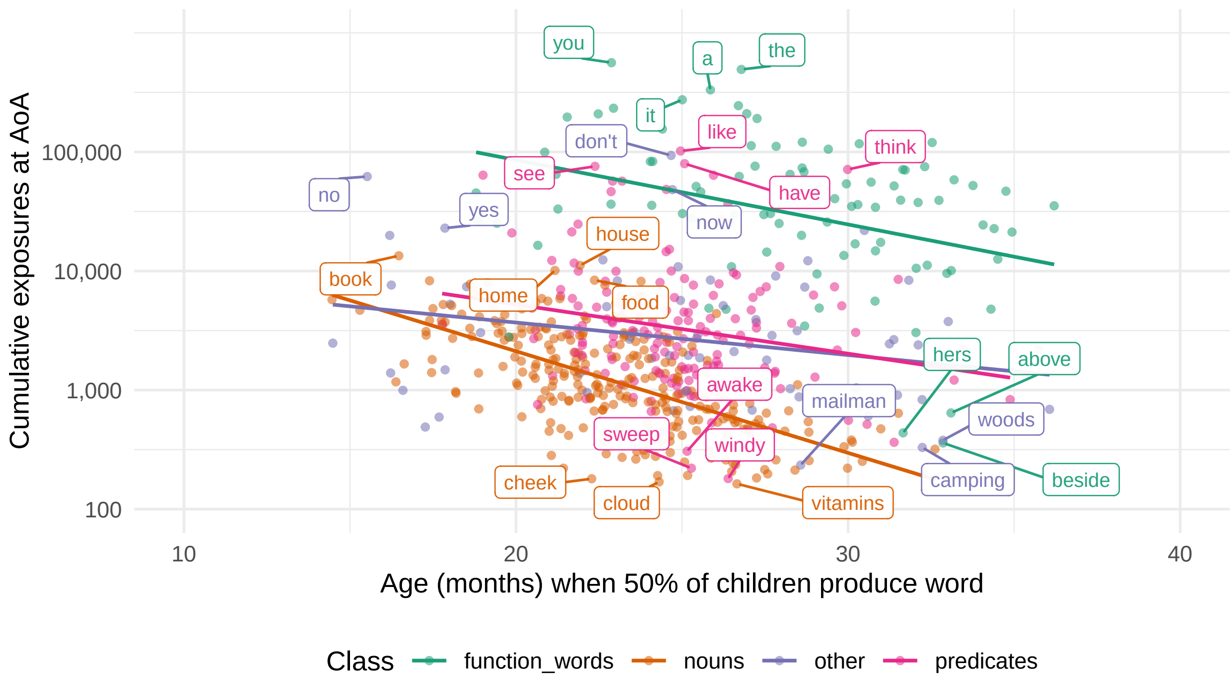

Figure 4

Large language models do not show acceleration in their vocabulary growth

Small-scale language models trained on developmentally-realistic corpora have become a useful tool for understanding early language development (**huebner2022?**; **futtrell2026?**; **frank2025?**). For our final experiments, we were interested in whether large language models show the same patterns of developmental change in efficiency or acceleration. We made use of work by Chang and Bergen (2022), who extracted per-word acquisition curves from language models. They computed logistic fits to the surprisal (negative log probability) of individual words across training, estimating

We can compare

(**evanston2023?**) investigated

We end by providing a quantitative comparison between within-child human learning and LLM scaling laws, showing that children's early language learning is strikingly different from LLM scaling: it shows both steeper scaling and greater variance in growth rate.

Discussion

What we did

Longitudinal growth models efficiency and acceleration. Two different paramters.

Where does acceleration come from? Could come from a variety of sources, including general maturation of memory and attention, specific changes in word recognition or processing (Fernald et al. 1998; **frank2026?**), the increasing sophistication of inferential strategies for word learning

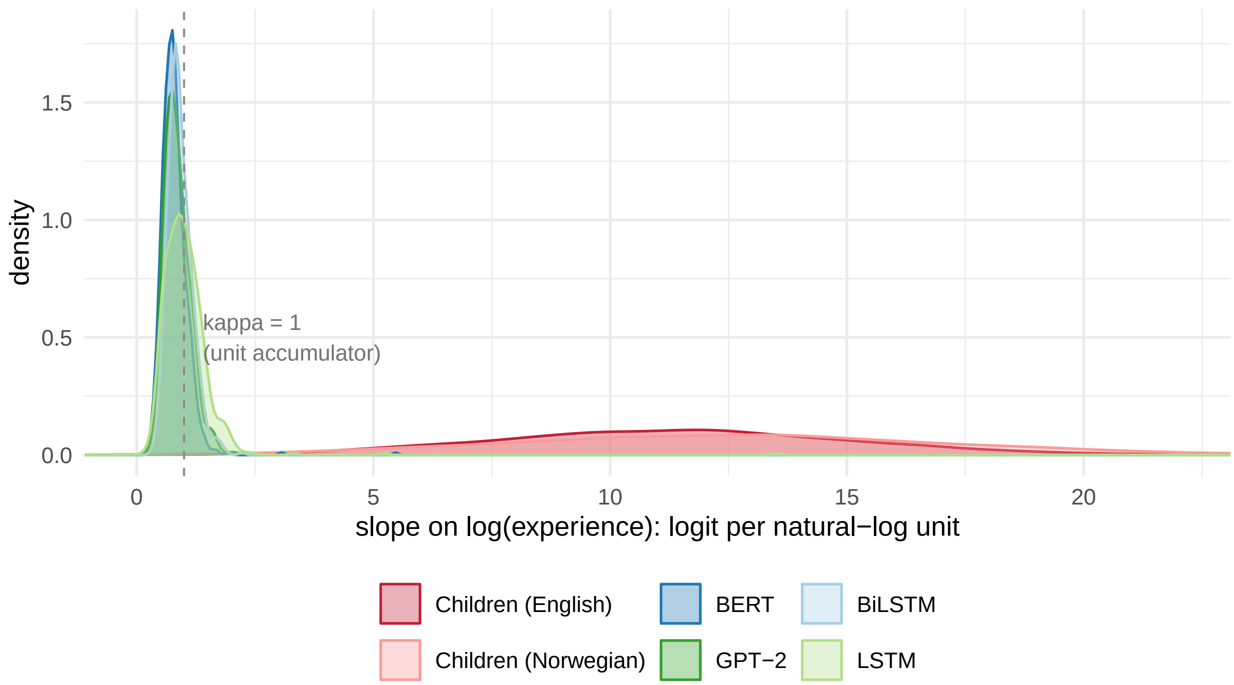

Figure 5: Comparison of acceleration slopes (kappa) estimated in our experiments for children in English and Norwegian, vs. LLM slopes per model estimated by Chang & Bergen (2022) and in our experiments.

[(**smith2002?**);Markman and Wachtel (1988);etc], or other sources.

Mitchell and McMurray (2009) prove that acceleration does not result from simply leverage of vocabulary –

Role of input: Coffey 2026 variance decomposition

Furthermore, how does the accumulation process relate to the rapid growth of vocabulary? On some models, accumulation by itself can produce an exponential "vocabulary explosion" while the child's own learning abilities remain constant. On other accounts, however, the child's own abilities change over development. These changes could be due to the accumulation of language allowing bootstrapping (e.g., via mutual excusivity or the shape bias), they could be through faster processing (Fernald rich get richer), or they could be via general maturation (e.g., better memory, faster processing). Or all of these.

Thus, modeling human learning requires positing other sources of variability and acceleration. Sources of variability could include from variation in interactional input quality, genetic variation between children, environmental richness not captured by input, etc. Sources of acceleration could include increases in processing efficiency, learning-to-learn including mutual exclusivity, shape bias, syntactic bootstrapping, as well as domain general changes in perception, processing speed, memory, attention or other aspects of cognition. The key to understanding children's data efficiency may thus lie in better understanding when and how children's learning accelerates with maturation and experience.

Network models and their predictions e.g. (**steyvers2005?**; **hills2009?**; Fourtassi, Bian, and Frank 2020)

Methods

Data and Code

Data were retrieved from Wordbank using the `wordbankr` package (Frank et al. 2017), childes-db using childesr (Sanchez et al. 2019), and peekbank using peekbankr (**zettersten2022?**). BabyView data are from release 2025.2, specifically those used by (**tan2026?**). SEEDLings data are downloaded from the materials provided by Egan-Dailey and Bergelson (2025).

Reproducible code for this paper is available at http://github.com/langcog/standard-model-2.

Models

Frequentist models

Initial models were fit with `glmer` (**glmer?**), with the following specifications:

- M1: `produces ~ offset(log_age) + (1 | item)`
- M2: `produces ~ 1 + log_age + (1 | item)`
- M3: `produces ~ 1 + log_age + (1 | item) + (1 | child)`
- M4: `produces ~ 1 + log_age + (1 | item) + (log_age | child)`

Linear variants reported in the supplement used linear age rather than log age.

Bayesian models

Bayesian models were fit using Stan (**stan?**).

Priors

Inference

Input

GPT-2 models

We trained XYZ

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Claude Code was used for literature review, code generation, visualization, and simulation infrastructure. All writing is original to the author.

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